

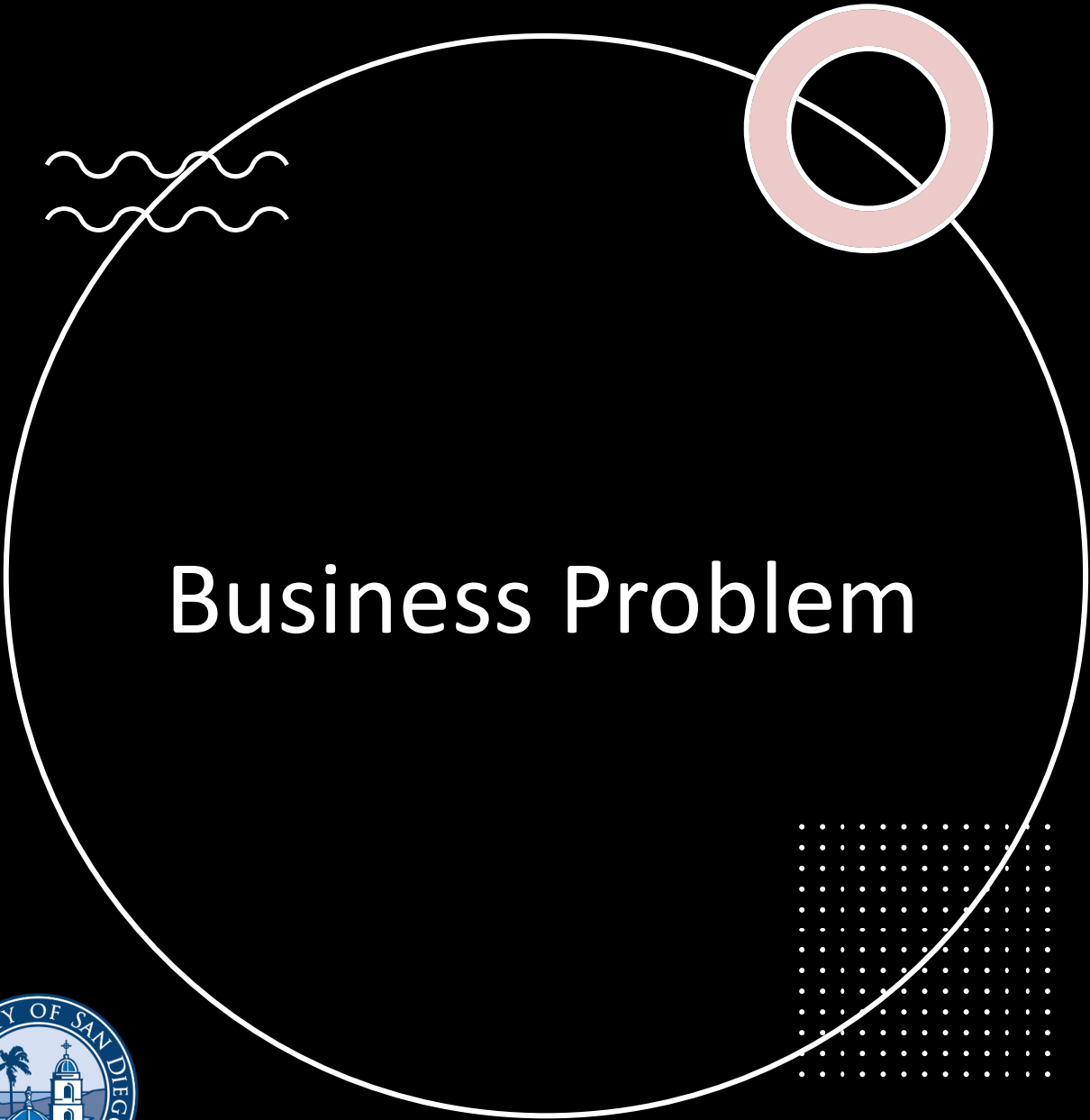
Imputation of Missing Chemical Concentration and CAS Registry Numbers Using MICE



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AAI 510 – Machine Learning Fundamentals





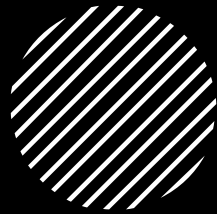
Business Problem

- Incomplete CAS Numbers and concentration values reduce regulatory compliance, hazard assessment quality, and chemical inventory accuracy.





Research Objectives



- Evaluate MICE for missing value recovery,
- Compare joint vs concentration-only imputation,
- Identify limitations of high-cardinality categorical targets.





Dataset Overview



16,703 chemical product records



Features: Trade Name, Chemical Name, CAS Number, Concentration



High-cardinality dataset with 1,800+ CAS values.





Methodology



Train/Test split (80/20)

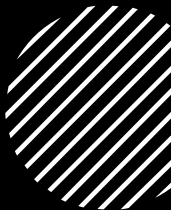
20% masking in test data

Ordinal encoding

MICE (Iterative Imputer)

50 iterations

Controlled evaluation protocol.





EDA & Data Preparation



Schema validation,

Completeness checks,

Categorical encoding,

Concentration distribution analysis,

Train-test segregation.



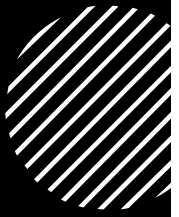
Model & Evaluation



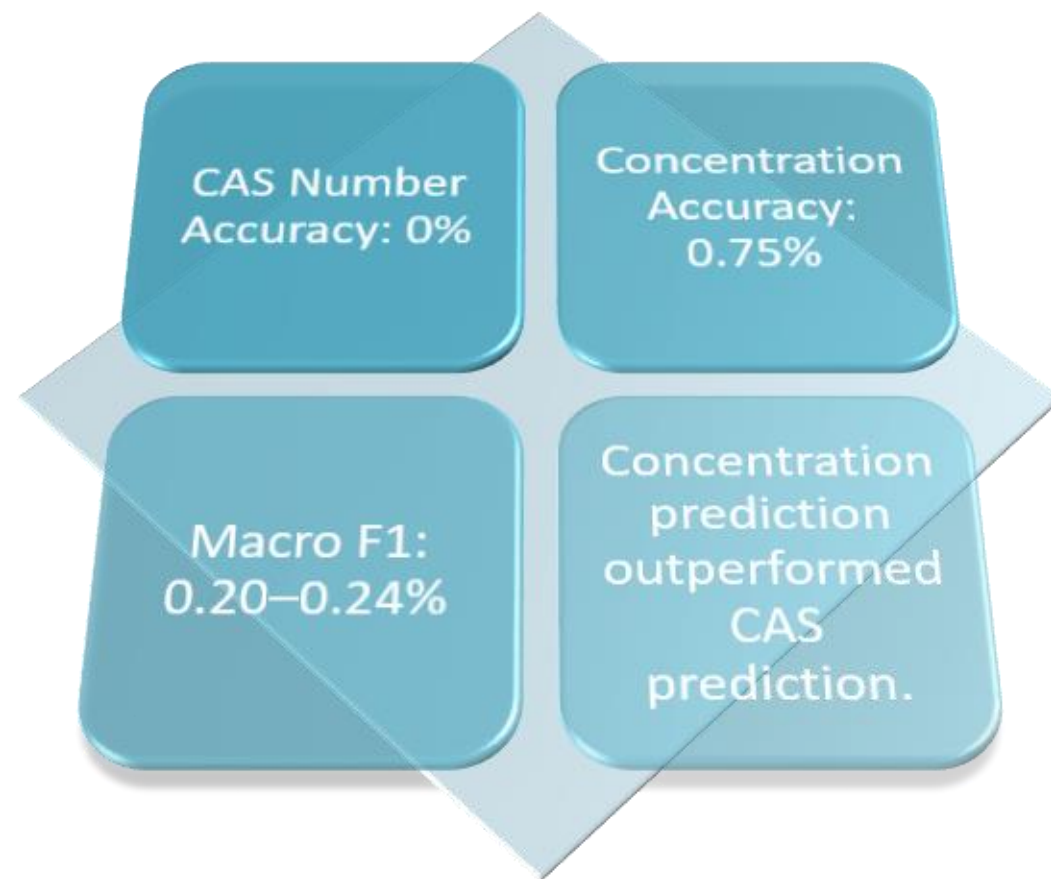
IterativeImputer (MICE)

Metrics: Accuracy and Macro F1

Evaluation only on masked values to ensure unbiased assessment.



Results





Business Impact & Deployment



Batch processing workflow
for chemical registries.

Can improve data quality
audits, compliance reviews,
and downstream analytics.

Deploy via API or scheduled
batch pipeline.

Conclusions & Future Work

- MICE provides a reproducible baseline.
- High-cardinality identifiers remain challenging.
- Future work: Random Forest, CatBoost, MissForest, Deep Learning approaches.

